Green University of Bangladesh

Department of Computer Science and Engineering (CSE)

Faculty of Sciences and Engineering

Semester: (Summer, Year:2026), B.Sc. in CSE (Day)

Lab Report NO : 1

Course Title: Artificial Intelligence Lab

Course Code: CSE 316 **Section: 241-D2**

Lab Experiment Name: Implementation of Basic Python Programs for Arithmetic Operations and List Processing.

Student Details

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Lab Date : **15-06-2026**
Submission Date : **22-06-2026**
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Lab Report Status

Marks: …………………………………………. **Signature:.............................**

Comments:.. **Date:...................................**

1. TITLE OF THE LAB REPORT EXPERIMENT

Implementation of Basic Python Programs for Arithmetic Operations and List Processing.

2. OBJECTIVES

- To understand the basic concepts of Python programming.
- To implement Python programs using loops, conditions, functions, lists, and tuples.
- To perform arithmetic operations and generate the Fibonacci series.
- To develop problem-solving and logical programming skills using Python.

3. PROCEDURE

- ❖ Open the Python IDE (Google Colab/Jupyter Notebook/VS Code).
- ❖ Write the required Python programs.
- ❖ Use appropriate loops, conditional statements, functions, lists, and tuples where necessary.
- ❖ Execute each program and provide the required inputs.
- ❖ Observe and verify the outputs.
- ❖ Debug any errors and rerun the programs if needed.
- ❖ Record the final outputs and results for analysis.

4. IMPLEMENTATION:

a) **<u>Sum of Odd and Even Numbers:</u>**

```
n = int(input("Total numbers: "))

odd_sum = 0
even_sum = 0

for i in range(n):
    num = int(input("Enter number: "))

    if num % 2 == 0:
        even_sum += num
    else:
        odd_sum += num

print("Sum of Even Numbers =", even_sum)
print("Sum of Odd Numbers =", odd_sum)
```

b) Sum Between 50 and 100 Divisible by 3 but Not by 5:

```
total = 0
for i in range(50, 101):
    if (i % 3 == 0) and (i % 5 != 0):
        total += i

print("Sum =", total)
```

c) Factorial of a Number:

```
num = int(input("Enter a number: "))
fact = 1

for i in range(1, num + 1):
    fact *= i

print("Factorial =", fact)
```

d) Fibonacci Series:

```
n = int(input("How many terms: "))

a = 0
b = 1
print("Fibonacci Series:")

for i in range(n):
    print(a, end=" ")
    c = a + b
    a = b
    b = c
```

e) Sum of Numbers Passed as Parameters:

```
def add(a, b, c):
    return a + b + c

result = add(10, 20, 30)
print("Sum =", result)
```

f) Reverse a Tuple:

```
t = (10, 20, 30, 40, 50)

reversed_tuple = t[::-1]

print("Original Tuple =", t)
print("Reversed Tuple =", reversed_tuple)
```

g) Swap Two Tuples:

```
t1 = (1, 2, 3)
t2 = (4, 5, 6)

print("Before Swapping:")
print("Tuple 1 =", t1)
print("Tuple 2 =", t2)

t1, t2 = t2, t1

print("\nAfter Swapping:")
print("Tuple 1 =", t1)
print("Tuple 2 =", t2)
```

h) Count Even and Odd Numbers in a List:

```
numbers = [10, 15, 20, 25, 30, 35, 40]

even_count = 0
odd_count = 0

for num in numbers:
    if num % 2 == 0:
        even_count += 1
    else:
        odd_count += 1

print("Even Numbers =", even_count)
print("Odd Numbers =", odd_count)
```

5. OUTPUT

```
Total numbers: 4
Enter number: 1
Enter number: 2
Enter number: 3
Enter number: 4
Sum of Even Numbers = 6
Sum of Odd Numbers = 4
```

Fig-1.1: Result of (a).

```
Sum = 1050
```

Fig-1.2: Result of (b).

```
Enter a number: 6
Factorial = 720
```

Fig-1.3: Result of (c).

```
How many terms: 6
Fibonacci Series:
0 1 1 2 3 5
```

Fig-1.4: Result of (d).

```
Sum = 60
```

Fig-1.5: Result of (e).

```
Original Tuple = (10, 20, 30, 40, 50)
Reversed Tuple = (50, 40, 30, 20, 10)
```

Fig-1.6: Result of (f).

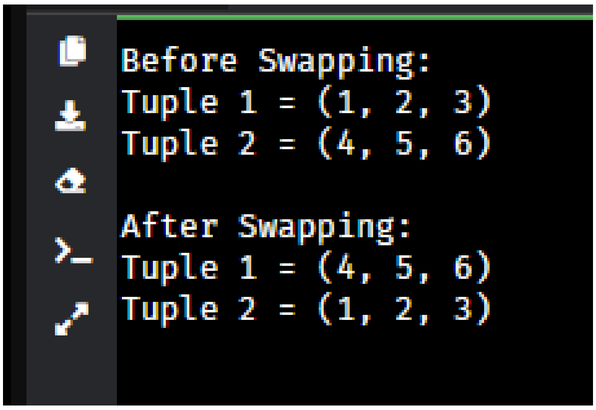

Fig-1.7: Result of (g).

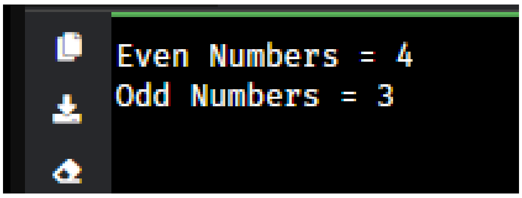

Fig-1.8: Result of (h).

6. ANALYSIS AND DISCUSSION

The programs were successfully implemented and executed using Python. Various programming concepts such as loops, conditional statements, functions, lists, and tuples were used. The outputs obtained were correct and matched the expected results. These programs helped in understanding basic Python programming techniques and problem-solving approaches.

7. SUMMARY

This lab experiment provided practical experience in Python programming. Different operations including arithmetic calculations, Fibonacci series generation, function implementation, tuple manipulation, and list processing were performed successfully. The experiment improved understanding of fundamental Python concepts and enhanced programming skills.

8. GITHUB REPOSITORY:

Link: https://github.com/pro382r/GUB-Academic/tree/main/AI-lab/lr1